

Partner Integration via Smobilpay Third Party API 3.0.5 – Workshop



Smobilpay Third Party (S3P) API

API Overview

API Protocol

The API is exposing a range of REST API return their data in JSON objects. Upstream communication in POST requests is transform as form data.

Communication Channel

The API is accessible only via SSL/HTTPS to ensure an encrypted transport level security during data exchange across all networks.

Authentication

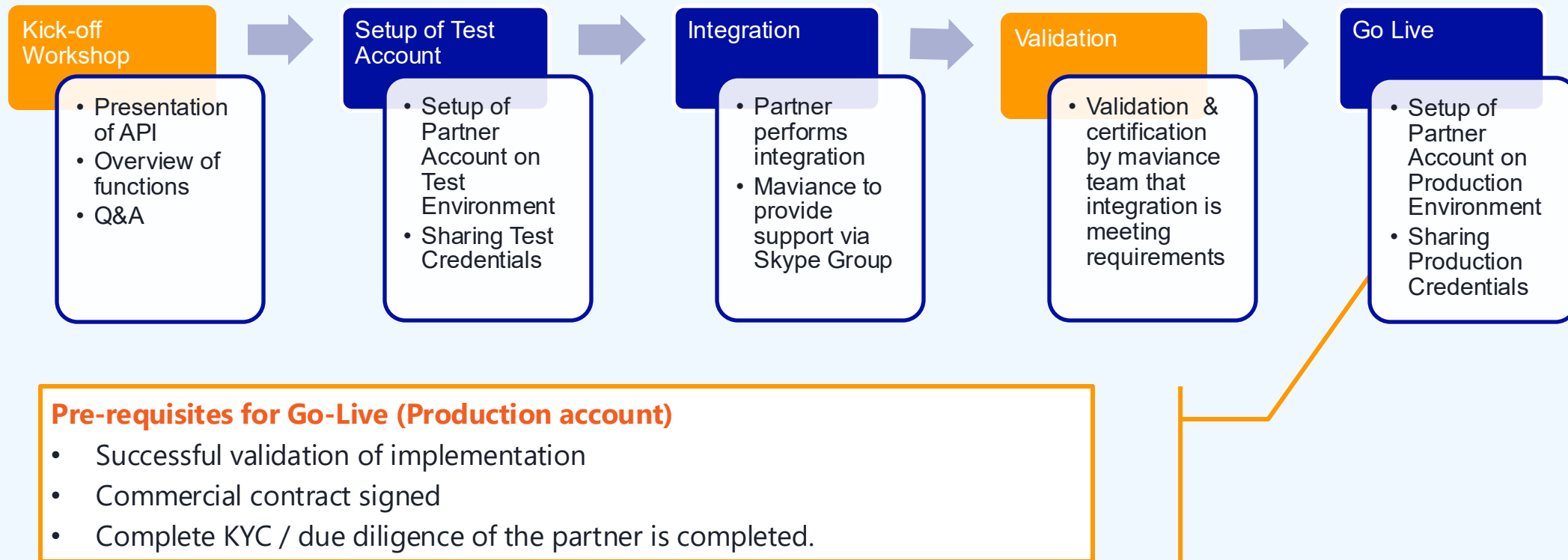
Each request must be authenticated according to our own **S3P Authorization Standard**. This method ensures that each request is secure, unique and protected against malicious or unintentional duplication and/or replay requests.



- Partners struggle most during the implementation of our Authorization Standard.
- We **highly recommend** to use one of our freely **available libraries** or **API Clients**.
- Where this is not possible, a descriptive step by step explanation for custom integrations is available [here](#)

Partner Integration Process

Overview



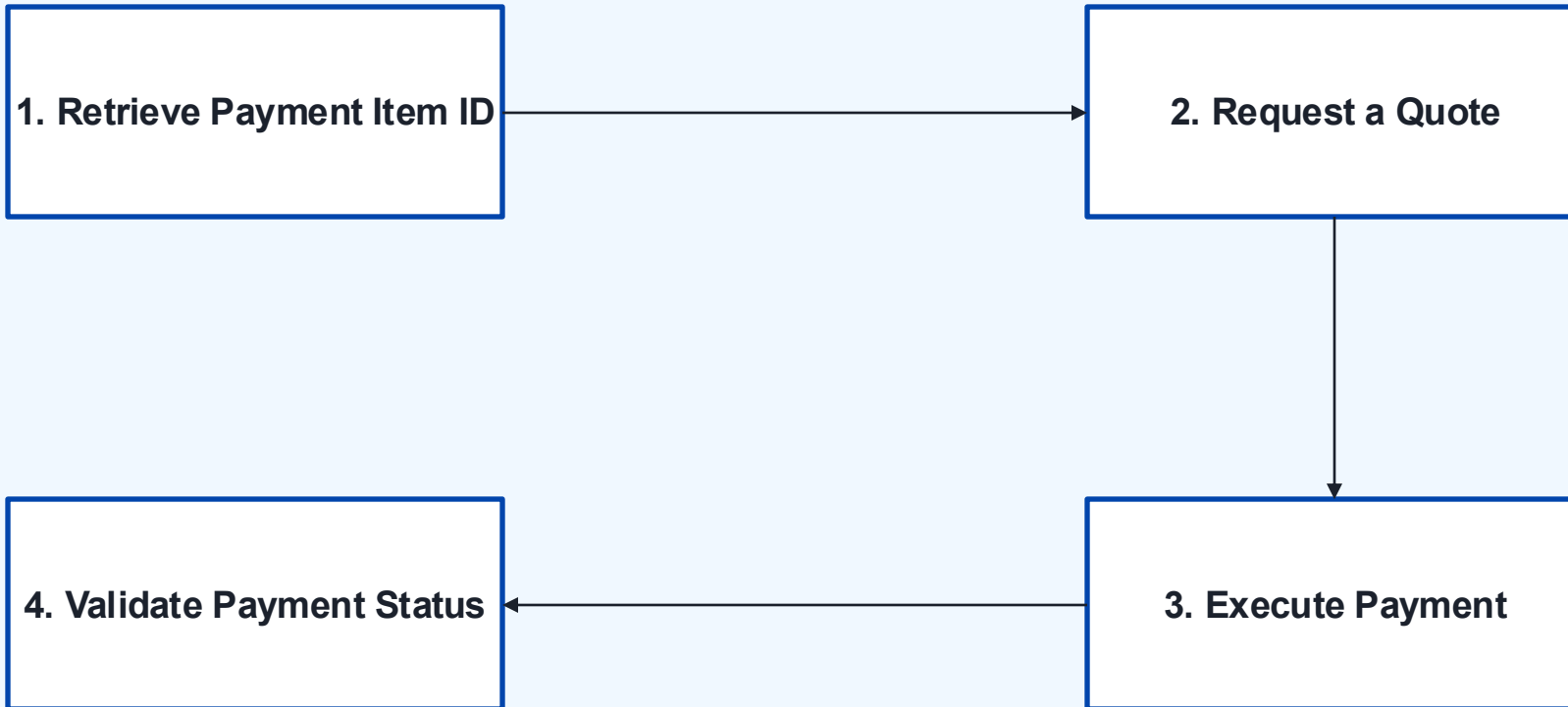
Smobilpay Third Party (S3P) API

Service Types

Service Types	Description of service type
Mobile Wallet (Cash-in)	• Credit Mobile Wallets & Mobile Money • Disburse into a customer's mobile wallets or mobile money account
Mobile Wallet (Cash-out)	• Debit Mobile Wallets & Mobile Money • Collect from a customer's mobile wallets or mobile money account • Receive and get paid online.
Top-up	• Top-up and airtime services
Product Purchase	• Purchase of products
Bill Payment	• Searchable bills (customer can search for and pay a specific bill) • Non-searchable bills (customer pays bills without searching)
Subscription	• Subscriptions are payments for payment items attached to regular payment cycles (e.g. Insurance Payments)
Voucher Purchase	• Vouchers are special products that return a digital code (printed or sent via email/SMS) to the customer

S3P API Call Flow

High Level Payment Flow



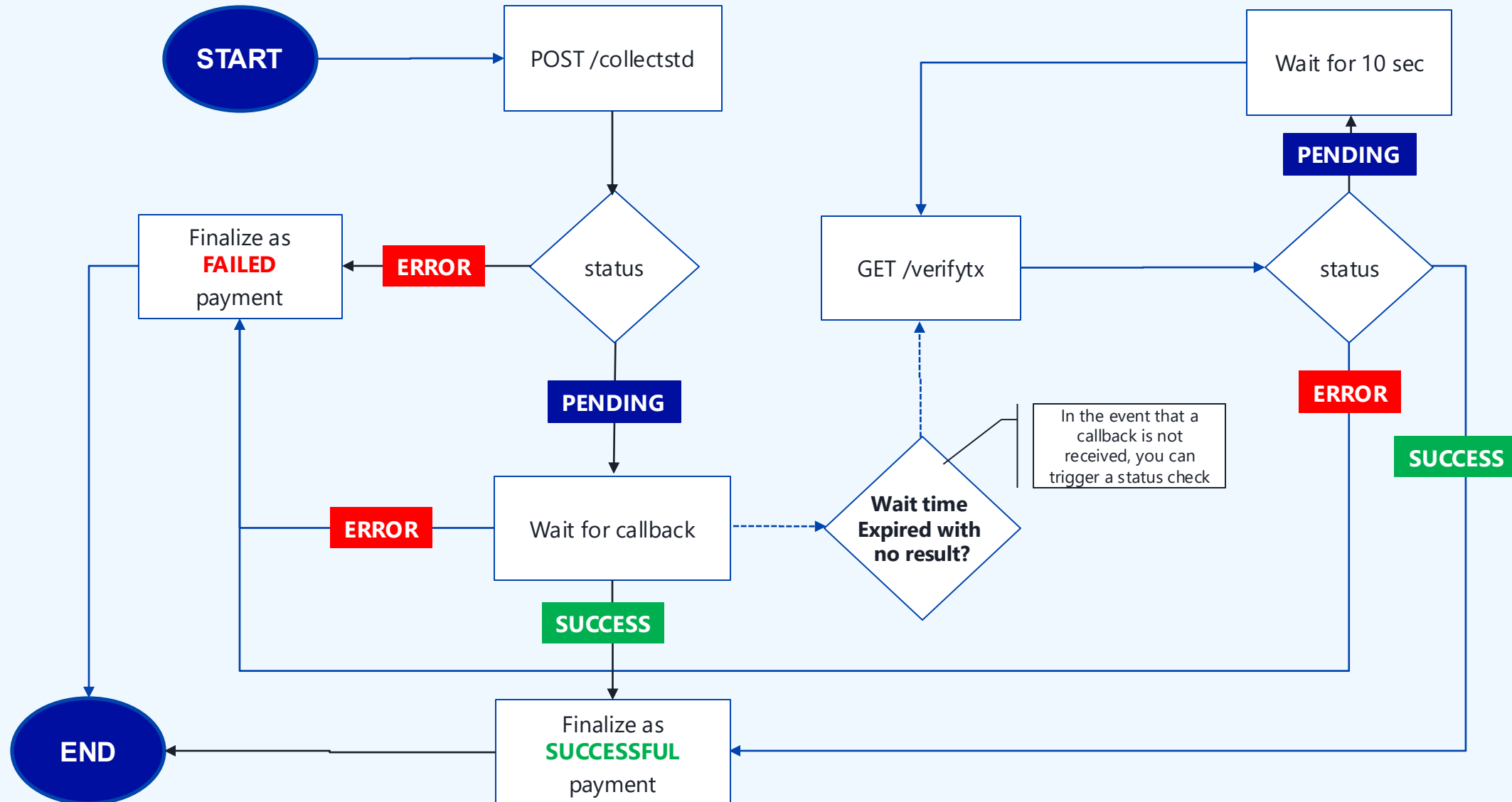
S3P API Payment Flow

Step #3 - Execute payment – Required data

Field	Type	Description
quoteld	REQUIRED	Quoteld from POST /quotestd endpoint
customerPhonenumber	REQUIRED	Phone number of customer for notification
customerEmailaddress	REQUIRED	Email address of customer for notification
customerName	DEPENDENT	Name of the customer (depends on service setting)
customerAddress	DEPENDENT	Address of the customer (depends on service setting)
serviceNumber	DEPENDENT	Destination / Account Number / Phonenumber / etc. (depends on service setting)
customerNumber	DEPENDENT	Customer Number / Account Reference / etc. (depends on service setting)
trid	OPTIONAL	Transaction id of the integrator

Verifying the status of a transaction

Payment Status Transition Flow



Verifying the status of a transaction

Using Callback – Recommended wait times for different services.

Service Types	Recommended WAIT TIME
Mobile Wallet (Cash-in)	• 20 seconds
Mobile Wallet (Cash-out)	• 2 Minutes In many cases the end-customers have to perform additional manual confirmation steps for 2 factor authorization of payment
Topup	• 20 seconds
Product Purchase	• 20 seconds
Bill Payment	• 20 seconds
Subscription	• 20 seconds
Voucher Purchase	• 20 seconds

Verifying the status of a transaction

Note:

1- You can only validate a transaction when you call the **GET /verifytx** endpoint until you get a final status.

2- Don't make assumptions on the final status of transactions without explicitly calling the endpoint **GET /verifytx** to get the final status.

3- We explicitly warn against the approach of setting a define time as you can see on the bottom of our API Documentation (**S3P API Concepts**)

) <https://apidocs.smobilpay.com/s3papi/S3P-API-Concepts.1578338258.html>

The screenshot shows a web browser window with the URL <https://apidocs.smobilpay.com/s3papi/S3P-API-Concepts.1578338258.html>. The page content includes a teal header, a back button, and a paragraph stating: "To provide tractability each payment transaction has a status indicating its state. The following statuses are supported:". Below this is a table with two columns: Status and Description.

Status	Description
PENDING	Payment has been sent to service provider, but confirmation has not yet been received. This is the default status.
ERRORED	Payment has failed
SUCCESS	Payment was properly processed and confirmed by service provider
REVERSED	Payment has been reversed (will only occur during reconciliation process)

Below the table, a red-bordered box contains the following text: "IMPORTANT: Only payments with status SUCCESS have been completely processed and confirmed. We recommend not to charge customers until the payment has this status." A red arrow points to this box.

Requirements for customer notifications

#	Scenario	Type	General	Collections (Cash-out)	Disbursement (Cash-in)	Bill Payment
1	Service number/bill number not found	Error	Mandatory	Mandatory	Mandatory	Mandatory
2	Informative message for customer to actively check confirm the transaction, even if there is no prompt (SMS or USSD) from Service Provider	Informative	N/A	Mandatory	N/A	N/A
3	Limit (daily, weekly or monthly) on recipient account has been reached	Error	N/A		Mandatory	N/A
4	Limit (daily, weekly or monthly) on Customer's account has been reached	Error	N/A	Mandatory	N/A	N/A
5	Invalid PIN or confirmation token	Error	N/A	Mandatory	N/A	N/A
6	Insufficient funds in the customer's account for this transaction	Error	N/A	Mandatory	N/A	N/A
7	Account is not enabled for this service (mobile money)	Error	N/A	Mandatory	Mandatory	N/A
8	Customer has denied approval for the collection request	Error	N/A	Mandatory	N/A	N/A
9	Customer failed to confirm the collection request	Error	N/A	Mandatory	N/A	N/A
10	The transaction was rejected because the amount is above the acceptable threshold	Error		Mandatory	Mandatory	Mandatory



Please note that during the verification phase, these notifications must be **validated before the production access is given.**

Errorcodes from the S3P API

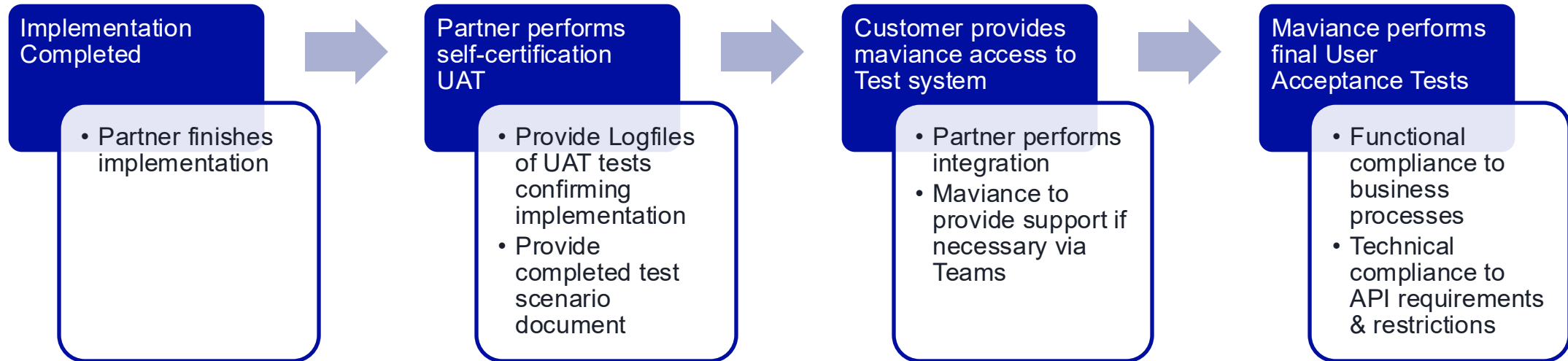
#	Scenario	Type	S3P Error Code
1	Service number/bill number not found	Error	42001
2	Informative message for customer to actively check confirm the transaction, even if there is no prompt (SMS or USSD) from Service Provider	Informative	N/A
3	Limit (daily, weekly or monthly) on recipient account has been reached	Error	703112
4	Limit (daily, weekly or monthly) on Customer's account has been reached	Error	703111
5	Invalid PIN or confirmation token	Error	703203
6	Insufficient funds in the customer's account for this transaction	Error	703108
7	Account is not enabled for this service (mobile money)	Error	703117
8	Customer has denied approval for the collection request	Error	703202
9	Customer failed to confirm the collection request	Error	703201
10	The transaction was rejected because the amount is above the acceptable threshold	Error	702103



Refer to the API documentation for the FULL set of error code ([LINK](#))

Certifying your integration

Overview of Certification Process



Certifying your integration

Integration Policies shall be verified during the validation process

During the certification process maviance will verify, amongst other criteria, the compliance to the following integration policies:

Endpoint	Reason	Recommended Call Frequency
GET /account	The account balance post payment is returned in the payment collection response There is no need to constantly re-request the account balance via this	On Demand
GET /verifytx	Transaction Status Endpoint must not be called excessively for the same payment. Some transactions take time to process in connected system.	Every 10 seconds per tx as long as status is PENDING. Switch to callback
Master data (except searches)	Master data does not change often and should be cached in the partner application. We recommend a daily data flush	Daily



- Please note that the production API is **monitored**
- Violations can result in accounts being **automatically suspended** without warning by the system

Smobilpay Third Party (S3P) API

Dynamic API & Service Identifiers

Dynamic API

As Smobilpay keeps expanding its network of supported integrations and service providers, it is necessary to offer an API that allows for this growth, without having to constantly extend and change the API to include new services.

To relay this dynamic nature, the S3P API provides access to service providers via **generic** configurations in form of a construct called **service**.

A **service** is essentially a collection of attributes that define business level meta information (name, category, merchant, logo, etc.) as well as technical **configuration** data (service type, required input data, labels, regex, denomination, etc.) to simplify client integration and allow for client-side pre-validation of input data. Each service has a **service id**, which uniquely identifies the service in Smobilpay.

This approach allows integrators (if they choose) to build a dynamic and generic implementation, based in **types** rather than a single payment provider / service. Smobilpay would ensure that new integrations of the same **type** follow the same flow as existing integrations.



The **service identifier** in the **test** system can be **different** than the one in the **production** system.

Service Type based Payment Flow

Each service is categorized into a specific **Service type** (e.g. Bill Payment, Cash in, Top-up, etc), which points to the kind of payment flow that needs to be followed in order to execute a payment for a particular service - as the information and data flow for e.g. a bill payment is different from a top-up.

Smobilpay Third Party (S3P) API

Dynamic API & Service Identifiers

Service Configuration

- Meta Information about service (names, labels, type, hints,...)
- Service level switches (customerNumber or serviceNumber) and related information (validation regex, labels)

RESPONSE SCHEMA: application/json

Array () [

serviceid

required

integer

Unique service Identifier. Use this value whenever "serviceid" is required in request parameters

merchant

required

string

Unique merchant code

title

required

string

Public name of service

description

required

string

Service description

category

required

string

Category of service

country

required

string

3 Characters

Country of operation (ISO 3166-1 alpha-3)

localCur

required

string

3 characters

Local currency of service. (Format: ISO 4217)

type

required

string

Enum: "SEARCHABLE_BILL" "NON_SEARCHABLE_BILL" "PRODUCT" "TOPUP" "SUBSCRIPTION" "CASHIN" "CASHOUT" "VOUCHER"

Type of service. This API will only provide services of the type

status

required

string

Enum: "Active" "Inactive"

Service availability status

isReqCustomerName

required

boolean

If set to true (1), the customers full name needs to be provided in the payment collection request.

isReqCustomerAddress

required

boolean

If set to true (1), the customers address needs to be provided in the payment collection request.

isReqCustomerNumber

required

boolean

If set to true (1), a customer number needs to be provided in the payment collection request. Customer number meaning is different for each service.

isReqServiceNumber

required

boolean

If set to true (1), a service number needs to be provided in the payment collection request. Service number meaning is different for each service.

labelCustomerNumber >

Array of objects (i18nText)

Label for customer number in multiple languages (if available) for this service.

labelServiceNumber >

Array of objects (i18nText)

Label for service number in multiple languages (if available) for this service.

isVerifiable

required

boolean

If set to true (1), then the service number provided for this service can be verified before making a payment request

validationMask

string

Optional mask for the service number entered during a payment for client side validations. All service numbers must comply to the mask in order to pass. The mask is a PCRE regular expression

hint >

Array of objects (i18nText)

Translation texts for the hint notes to be displayed to the customer for this service.

denomination

integer

Service payment denomination. The payment amount must be a multiple of the denomination value. Example:

Denomination	Amount	Valid
1	100	true
5	100	true
200	100	false
50	60	false

Communication channels

During Integration Phase

- To speed up integration, a dedicated channel shall be created for working together with your team.
- Use the channels below
 - Teams channel
 - Email: integration-support@cm.maviance.com

During Production Phase

The MS Teams channels created for integration shall **NO LONGER** be supported.

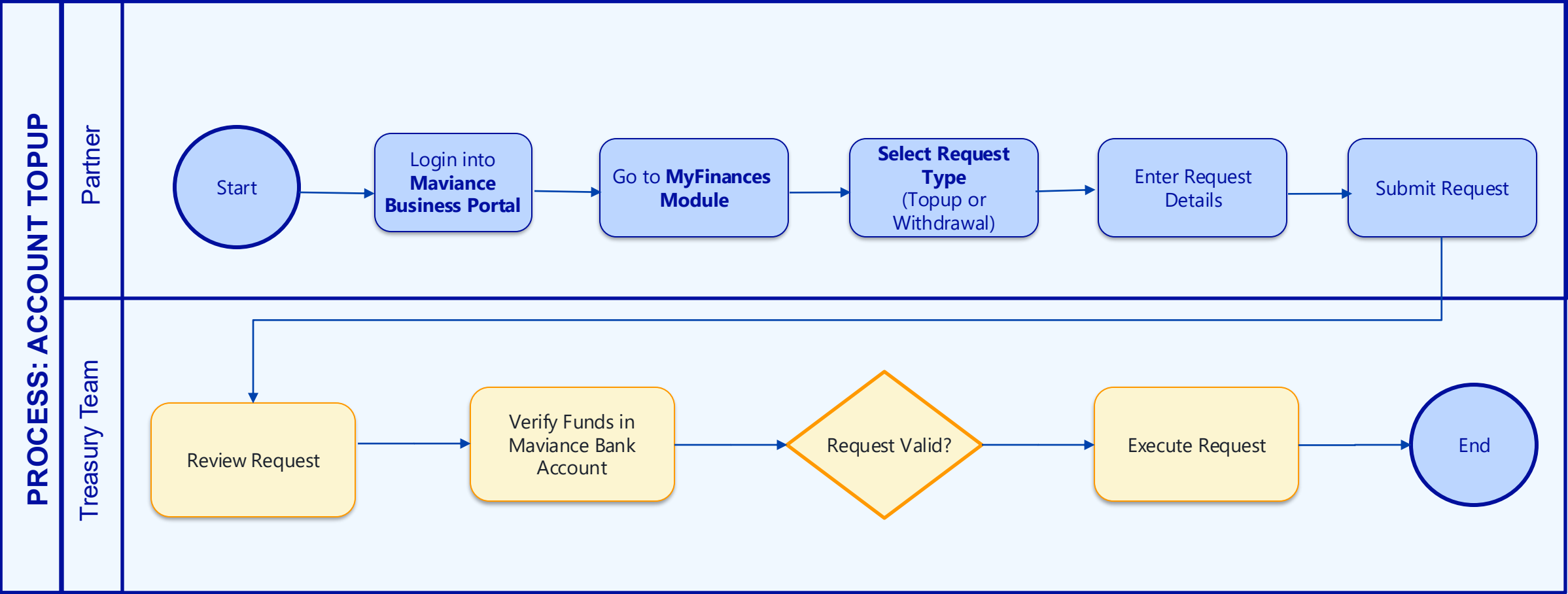
All support questions should be sent to the central channels

Service Desk: <https://support.cm.smobilpay.com>

Email: support@cm.smobilpay.com

Submit Account Topup & Withdrawal Requests via Maviance Business Portal

 **Link to Maviance Business Portal:** <https://pay.maviance.com>



Thank You!

MAVIANCE PLC

Avenue Charles de Gaulle
NFC Bank Building, Bonanjo
P.O Box 8520, Douala – Cameroon

contact@cm.maviance.com
www.maviance.cm

Submit Payout Requests and **ONLY** via Maviance Business Portal

 **Link** to Maviance Business Portal: <https://pay.maviance.com>

